# DroneVault – A Secure Drone Operations & Analytics Platform


VJTI

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## PROBLEM STATEMENT:

The rapid growth of consumer drones has created two major challenges: lack of a private, offline tool to analyze flight data, and increasing security risks from unauthorized drones. Existing solutions depend on cloud services, raising data privacy concerns, while traditional counter-drone methods disrupt nearby networks. Therefore, there is a need for a secure, platform that enables efficient drone data analysis and safe monitoring without relying on external systems.

## ABSTRACT:


DroneVault is a privacy-first platform designed for drone forensics and security. It allows the users to import and analyze DJI and Litchi flight logs, store telemetry data in a local database, and visualize flights through interactive 3D replays, charts, and battery health reports. Our platform also supports exporting data for reporting and compliance. In addition, it includes a security module that monitors nearby Wi-Fi networks, identifies potential rogue drone threats, and selectively blocks the unauthorized drones using deauthentication without affecting trusted networks.


## APPLICATIONS:

- **Security & Surveillance** – Detect unauthorized drones in restricted areas.
- **Critical Infrastructure Protection** – Prevent UAV threats near sensitive facilities like power grids.
- **Fleet Management** – Track logs, battery health, and for UAV maintenance.
- **Incident Analysis** – Replay 3D flights for investigation.
- **Academic Research** – Study Wi-Fi protocols, drone failsafe behaviours, and battery degradation patterns.

### Tool Comparison

| | DroneVault | Cloud-based Logbook | Flight Reader |
|---|---|---|---|
| Data Privacy | 100% Local | Cloud Upload | 100% Local |
| Wi-Fi Security | Detects & Blocks Rogue Drones | ✗ | ✗ |
| Flight Replay | 3D view | 2D view | 3D view (Limited) |
| Import Support | Local folder | DJI cloud | DJI or local ✓ |
| Price | Free Forever | $6.99/month | $79 USD ✓ |

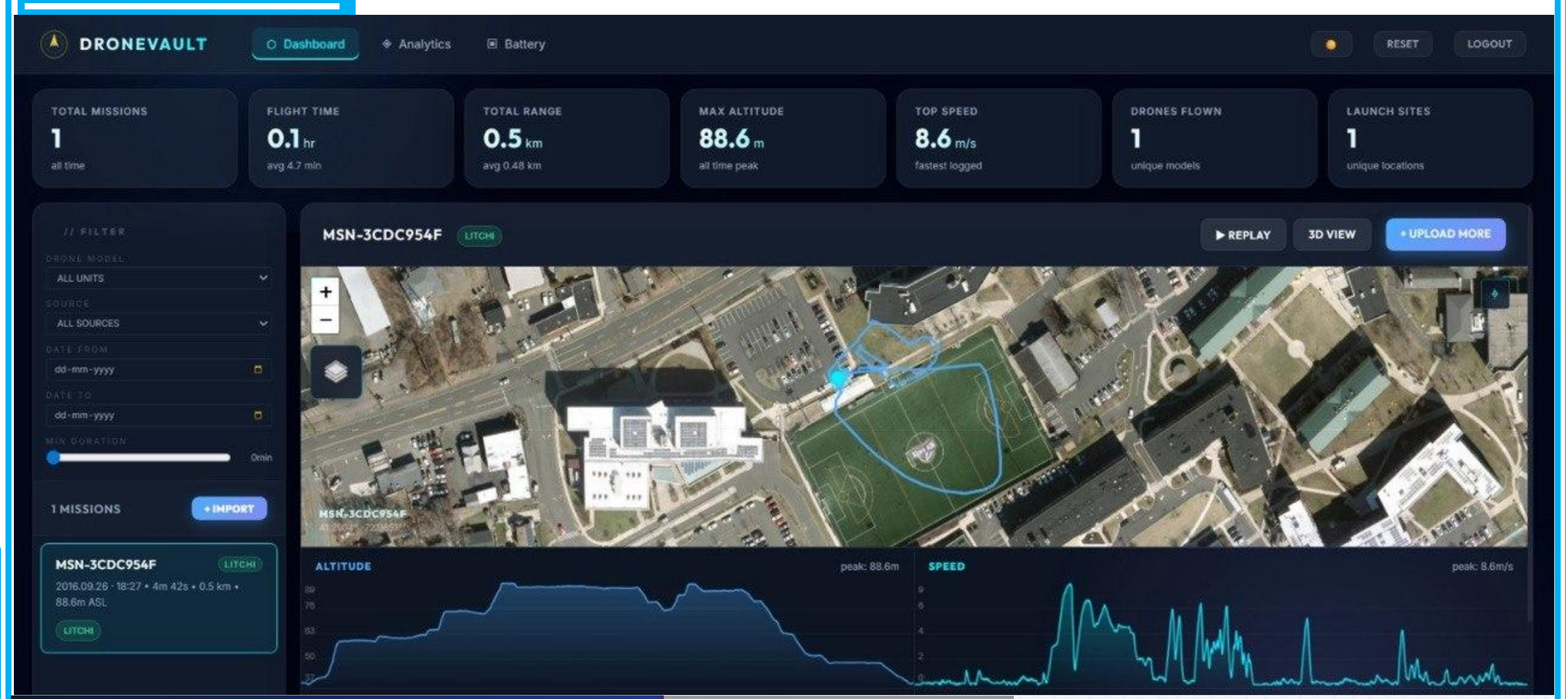
## CONCLUSIONS:

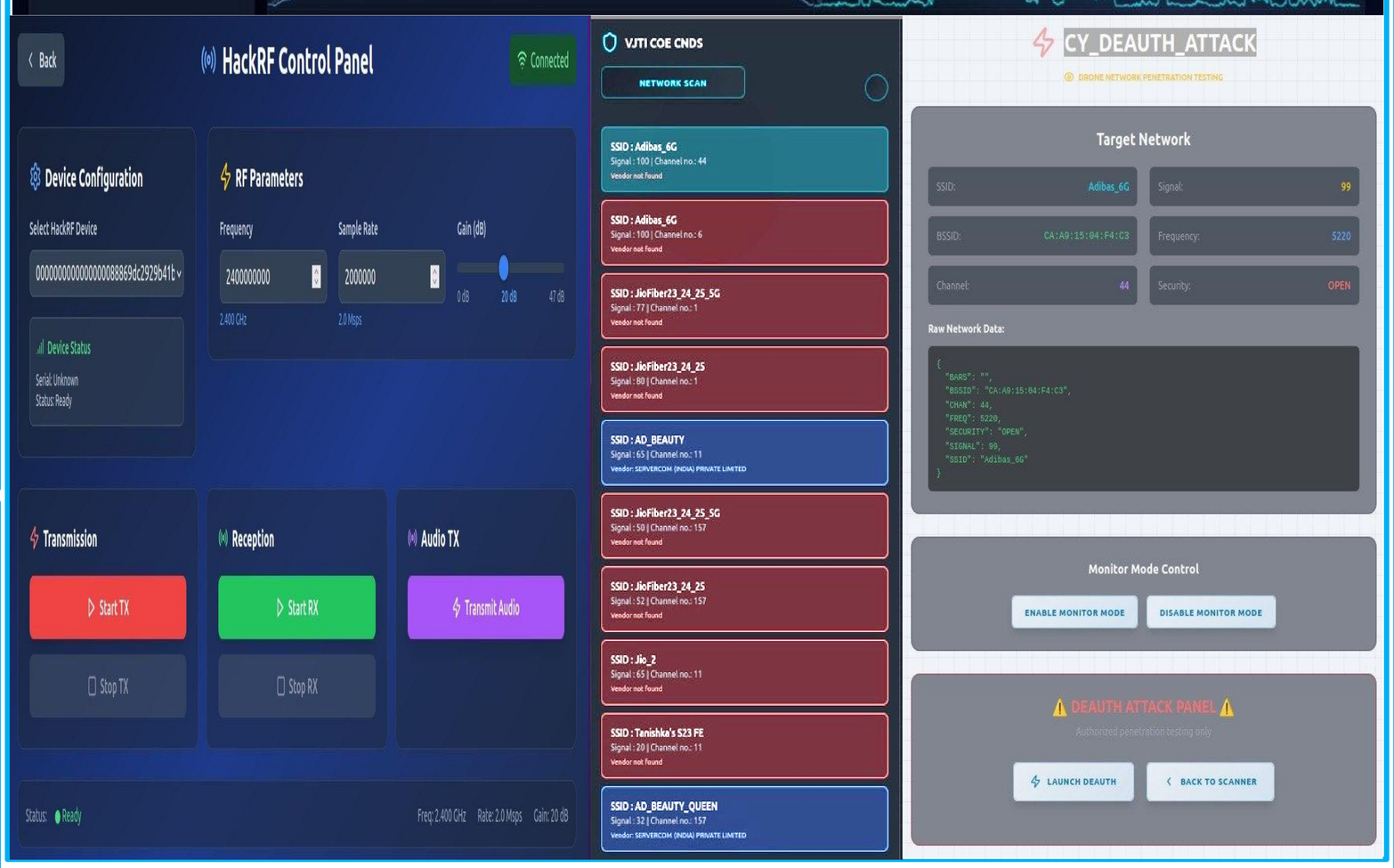
Our project combines drone forensics and security in a local platform. It analyzes flight logs and detects unauthorized drones while keeping all data private. Built with modern technologies, it is a scalable and deployable solution for real-world use.

**Future Scope:**

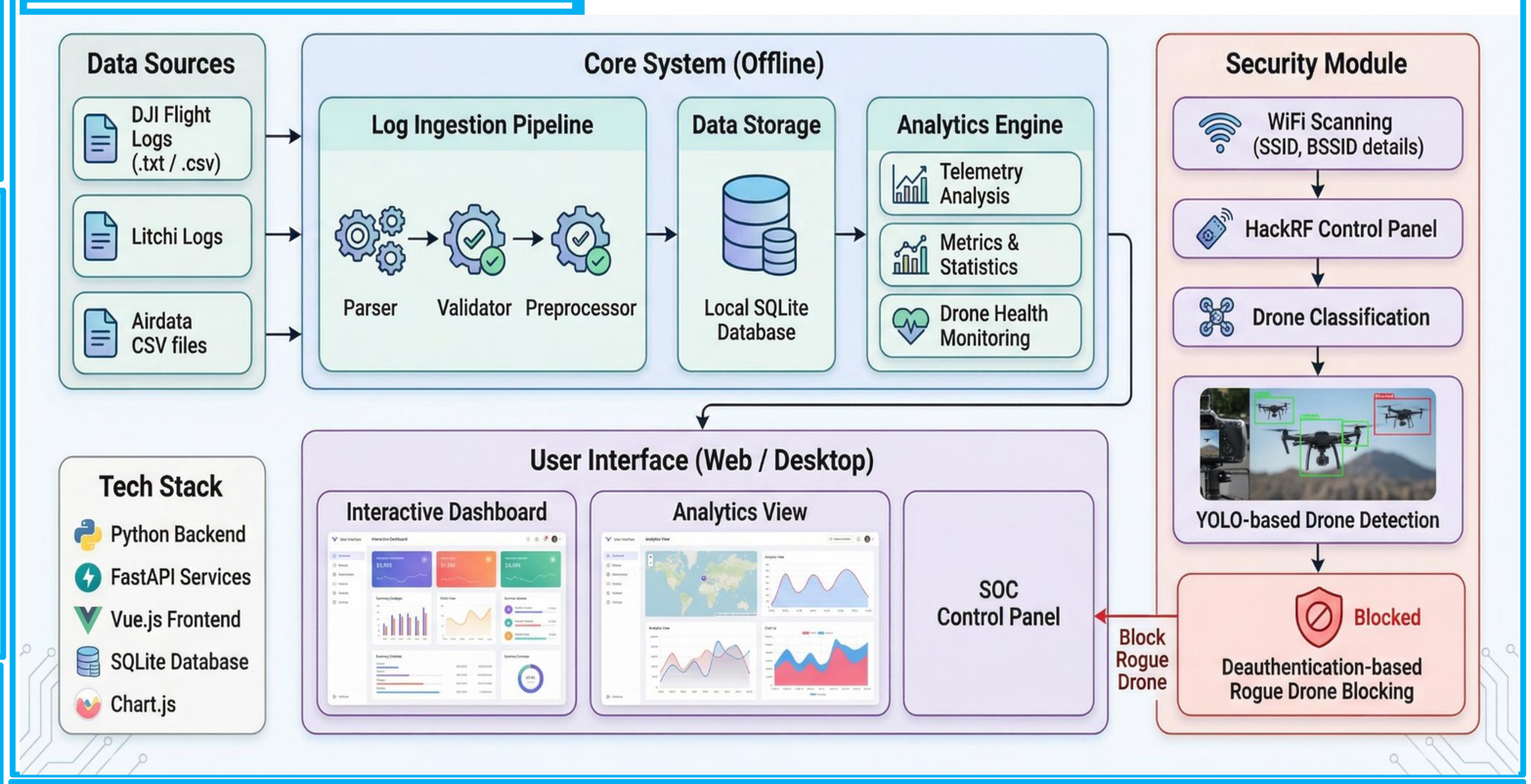
AI/ML-based forensic analysis, and advanced UAVs threat detection and mitigation.

## RESULTS:




## ARCHITECTURE:

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